

● MEDIUM

● INFORMATION AND IDEAS

Information and Ideas

02

10 questions · ~15 min

Q1

INFORMATION AND IDEAS

The following text is adapted from Jack London's 1903 novel *The Call of the Wild*. Buck is a sled dog living with John Thornton in Yukon, Canada.

Thornton alone held [Buck]. The rest of mankind was as nothing. Chance travellers might praise or pet him; but he was cold under it all, and from a too demonstrative man he would get up and walk away. When Thornton's partners, Hans and Pete, arrived on the long-expected raft, Buck refused to notice them till he learned they were close to Thornton; after that he tolerated them in a passive sort of way, accepting favors from them as though he favored them by accepting.

Which choice best states the main idea of the text?

- A. Buck has become less social since he began living with Thornton.
- B. Buck mistrusts humans and does his best to avoid them.
- C. Buck has been especially well liked by most of Thornton's friends.
- D. Buck holds Thornton in higher regard than any other person.

Many mosquito repellents contain natural components that work by activating multiple odor receptors on mosquitoes' antennae. As the insects develop resistance, new repellents are needed. Ke Dong and her team found that EBF, a molecular component of a chrysanthemum-flower extract, can repel mosquitoes by activating just one odor receptor—and this receptor, Or31, is present in all mosquito species known to carry diseases. Therefore, the researchers suggest that in developing new repellents, it would be most useful to _____ blank

Which choice most logically completes the text?

- A. identify molecular components similar to EBF that target the activation of Or31 receptors.
- B. investigate alternative methods for extracting EBF molecules from chrysanthemums.
- C. verify the precise locations of Or31 and other odor receptors on mosquitoes' antennae.
- D. determine the maximum number of different odor receptors that can be activated by a single molecule.

Characteristics of Five Recently Discovered Gas Exoplanets

Exoplanet designation	Mass (Jupiters)	Radius (Jupiters)	Orbital period (days)	Distance from the Sun (parsecs)
TOI-640 b	0.88	1.771	5.003	340
TOI-1601 b	0.99	1.239	5.331	336
TOI-628 b	6.33	1.060	3.409	178
TOI-1478 b	0.85	1.060	10.180	153
TOI-1333 b	2.37	1.396	4.720	200

“Hot Jupiters” are gas planets that have a mass of at least 0.25 Jupiters (meaning that their mass is at least 25% of that of Jupiter) and an orbital period of less than 10 days (meaning that they complete one orbit around their star in less than 10 days), while “warm Jupiters” are gas planets that meet the same mass criterion but have orbital periods of more than 10 days. In 2021, Michigan State University astronomer Joseph Rodriguez and colleagues announced the discovery of five new gas exoplanets and asserted that four are hot Jupiters and one is a warm Jupiter.

Which choice best describes data from the table that support Rodriguez and colleagues’ assertion?

- A. None of the planets have an orbital period of more than 10 days, and TOI-628 b has a mass of 6.33 Jupiters.
- B. TOI-1478 b has an orbital period of 153 days, and the masses of all the planets range from 0.85 to 6.33 Jupiters.

- C. All the planets have a radius between 1.060 and 1.771 Jupiters, and only TOI-1333 b has an orbital period of more than 10 days.
- D. Each of the planets has a mass greater than 0.25 Jupiters, and all except for TOI-1478 b have an orbital period of less than 10 days.

Q4

INFORMATION AND IDEAS

Historians point to the rule of the Piast dynasty as crucial to the formation of the Polish state. However, some differentiate between members of the dynasty like Mieszko II Lambert, who ruled as king from 1025 to 1031 CE, and less well-documented figures like Siemomysł, who is said to have ruled in the 10th century but whose historical actuality is disputed. Siemomysł appears in the *Gesta principum Polonorum*, a chronicle of medieval Polish history written between 1112 and 1118. However, the chronicle's documentation of Siemomysł relies on oral tradition, unlike its records of later rulers.

According to the text, what is a difference between how historians view Siemomysł and how they view Mieszko II Lambert?

- A. Historians agree that Mieszko II Lambert existed, but disagree about whether Siemomysł existed.
- B. Historians believe that the *Gesta principum Polonorum* provides more evidence for Siemomysł's existence than it does for Mieszko II Lambert's existence.
- C. Historians agree that Siemomysł ruled Poland much later than Mieszko II Lambert.
- D. Historians find the orally transmitted stories affirming the existence of Mieszko II Lambert to be more convincing than similar stories about Siemomysł.

Violins made by Antonio Stradivari and other craftspeople in the sixteenth to eighteenth centuries in Cremona, Italy, produce a sound that is considered superior to that of modern stringed instruments. Some experts have claimed that the type of wood used to create Cremonese violins is responsible for their prized sound, but modern and Cremonese violins are made of the same kinds of wood: maple and spruce. New analysis, however, has revealed unique indications that the wood in the older violins was chemically treated by the makers, leading researchers to suggest that _____ blank

Which choice most logically completes the text?

- A. Cremonese violins probably were not considered superior to other instruments at the time they were made.
- B. the sound quality of Cremonese violins results in part from a method the craftspeople used to alter the wood.
- C. if modern violins were made of a wood other than maple or spruce, they likely would sound as good as Cremonese violins.
- D. the current process of making violins is the same process that was used centuries ago by Cremonese craftspeople.

Five of the Responses to Survey about Actions to Conserve Energy

Action	Action category	Percentage of respondents selecting action (%)
Use efficient cars/hybrids	efficiency	2.8
Change thermostat setting	curtailment	6.3
Use bike or public transportation instead of car	curtailment	12.9
Use efficient light bulbs	efficiency	3.6
Turn off lights	curtailment	19.6

In a survey of public perceptions of energy use, researcher Shahzeen Attari and her team asked respondents to name the most effective action ordinary people can take to conserve energy. The team categorized each action as either an efficiency or a curtailment and found that respondents tended to name curtailments more often than they did efficiencies. For example, 19.6% of respondents stated that the most effective way to conserve energy is to turn off the lights, while only _____blank

Which choice most effectively uses data from the table to complete the text?

- A. 6.3% of respondents said it was most effective to use efficient cars or hybrids.
- B. 2.8% of respondents said it was most effective to change the thermostat setting.

- C. 12.9% of respondents said it was most effective to use a bike or public transportation.
- D. 3.6% of respondents said it was most effective to use efficient light bulbs.

Q7

INFORMATION AND IDEAS

NASA's Aspera mission, led by Carlos Vargas, will investigate the circumgalactic medium (CGM), the huge swaths of low-density gas that fill and surround galaxies. Specifically, the team will focus on portions of the gas that exist in a "warm-hot" phase: these portions haven't previously been observable but are thought to fuel new star formation and hold most of the mass that makes up a galaxy. Using a telescope capable of revealing these parts of the CGM, the Aspera mission should help answer long-standing questions about how galaxies emerge, change, and even interact.

Which choice best states the main idea of the text?

- A. As the leader of NASA's Aspera mission, Vargas will be the first person to investigate the makeup of the CGM.
- B. Although galaxies that are surrounded by the CGM have been studied, researchers have been unable to directly observe low-density gas in the CGM in the "warm-hot" phase.
- C. Researchers don't yet have a complete understanding of the process of galaxy evolution but have raised the possibility that galaxies interact with each other at times.
- D. The Aspera mission is expected to produce the first direct observations of CGM gas in the "warm-hot" phase, which likely has an important role in the evolution of galaxies.

Narwhals are shy whales that live in the remote Arctic Ocean. Some of them have a long tusk, like a unicorn horn, with sensitive nerves. Narwhals are known for this tusk, but many actually don't have one and its purpose is unknown. One group of scientists came up with a possible purpose in 2014. The scientists suggested that the tusk may help narwhals determine when water around them is likely to start freezing and become dangerous for them. Marine biologist Kristin Laidre disagrees with that idea, though. She reasons that if the narwhal's tusk serves such an important purpose, then it's most likely that _____ blank

Which choice most logically completes the text?

- A. some narwhals would seek a new habitat.
- B. fewer marine animals would also have tusks.
- C. more narwhals would have a tusk.
- D. narwhals would become less shy over time.

Sense and Sensibility is an 1811 novel by Jane Austen. In the novel, Austen describes Marianne Dashwood's ability to persuade others of the rightness of her artistic judgments, as is evident when Marianne visits with John Willoughby, a potential suitor: _____ blank

Which quotation from *Sense and Sensibility* most effectively illustrates the claim?

- A. "Above all, when she heard him declare, that of music and dancing he was passionately fond, she gave him such a look of approbation as secured the largest share of his discourse to herself for the rest of his stay."
- B. "Their taste was strikingly alike. The same books, the same passages were idolized by each—or if any difference appeared, any objection arose, it lasted no longer than till the force of her arguments and the brightness of her eyes could be displayed."
- C. "It was only necessary to mention any favourite amusement to engage her to talk. She could not be silent when such points were introduced, and she had neither shyness nor reserve in their discussion."
- D. "They speedily discovered that their enjoyment of dancing and music was mutual, and that it arose from a general conformity of judgment in all that related to either. Encouraged by this to a further examination of his opinions, she proceeded to question him on the subject of books."

Recently, scientists looked at data collected by NASA's InSight lander to learn more about seismic activity on Mars, known as marsquakes. The data show that the marsquakes all started from the same location on the planet. This discovery was surprising to scientists, as they expected that the marsquakes would originate from all over the planet because of the cooling of the planet's surface. Now, scientists believe that there could be areas of active magma flows deep beneath the planet's surface that trigger the marsquakes.

According to the text, what was surprising to scientists studying the seismic activity data from NASA's InSight lander?

- A. The surface temperature of Mars has been rising.
- B. There were different types of seismic waves causing marsquakes.
- C. NASA's InSight lander collected less data than scientists had expected.
- D. All the marsquakes started from the same location on the planet.